PITCH DECK · MLH DUCKTOBERFEST · TEAM CROWTECH

VEYRA

The security perimeter for AI agents.
Prompt injection stops here.

github.com/Ducktoberfest/team-crowtech



THE TEAM

Team Crowtech

Najaf Guluyev

Full-Stack Developer

Developed the API integration, user dashboard, frontend experience, and Vercel deployment.

Sonakhanim Dadashova

AI & Security Engineer

Developed the AI-powered threat detection engine, prompt-injection classification, risk scoring, and security policies.

Built for MLH Ducktoberfest · Open source under the MIT license

THE PROBLEM

AI agents act on every word they read

Chat, email, tickets, web pages and PDFs all flow into models that can call tools. One hidden instructaion is enough to take control.

HOW ATTACKERS GET IN

Five ways attackers hijack an agent

Instruction override	Commands the model to ignore its system directives
Jailbreak personas	DAN or Developer Mode coerce unfiltered responses
Data exfiltration	System prompts, API keys and environment variables leak
Dangerous actions	Shell commands or SQL drops forced through tool calls
Indirect injection	Hidden commands smuggled inside innocent-looking documents

THE SOLUTION

A policy-enforced gateway in front of every downstream LLM

Deterministic rules

Open-weight AI audit

Security policy engine

HOW IT WORKS

Three stages, one decision

STAGE 1

Deterministic rules

- Unicode and homoglyph normalization
- Zero-width space stripping
- Critical regex signatures

STAGE 2

Open-weight model audit

- Ollama endpoint running Gemma 4:31B
- Structured JSON taxonomy output
- Offline mock mode for demos

STAGE 3

Security policy engine

- Composite risk score from 0.0 to 1.0
- Configurable BLOCK and QUARANTINE thresholds
- Conservative fail-closed handler

POLICY ENGINE

Three outcomes for every input

ALLOW

< 0.35

Clean input is dispatched to the downstream agent context.

QUARANTINE

0.35 – 0.70

Borderline or indirect input is isolated for human review.

BLOCK

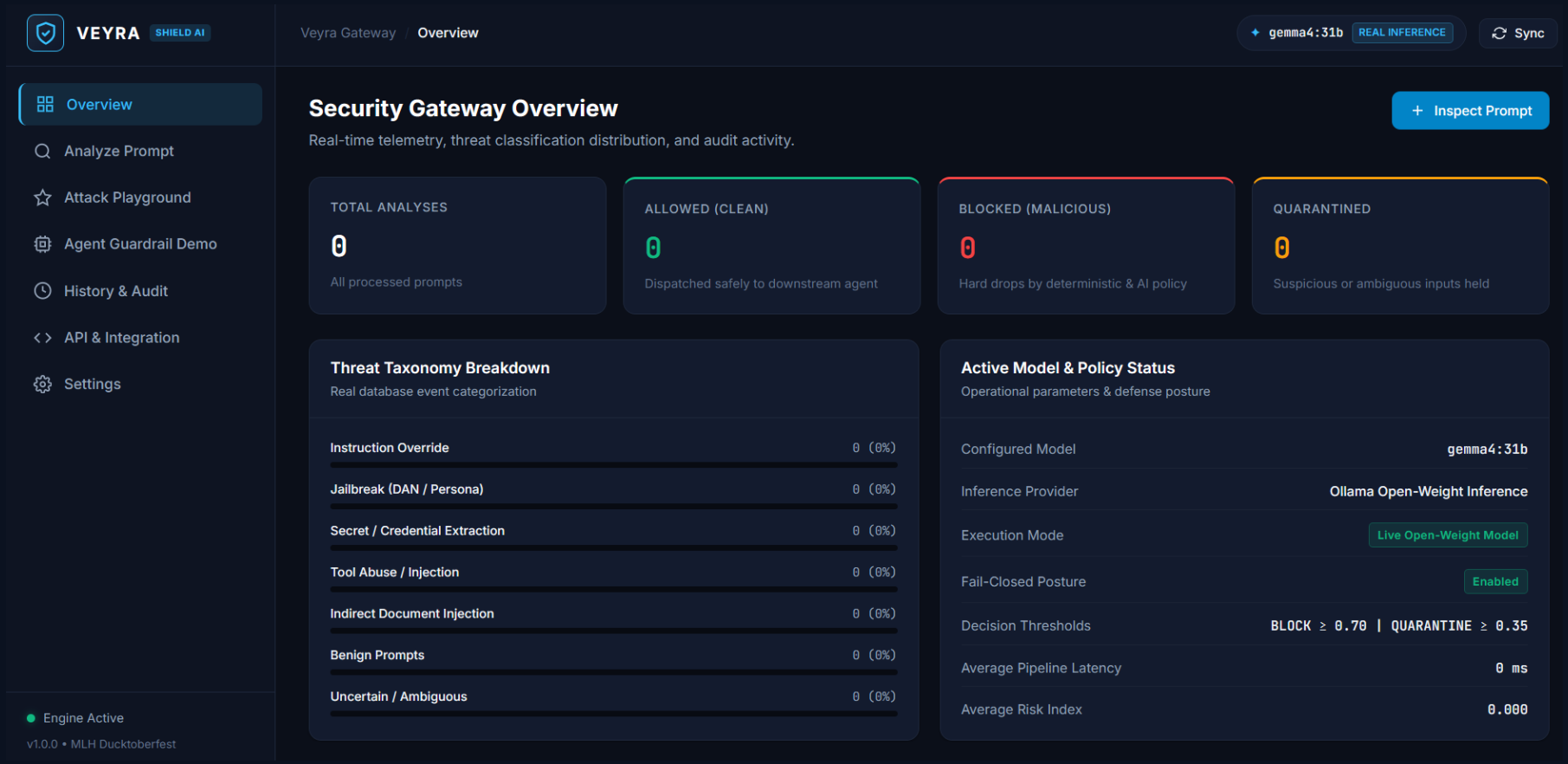
≥ 0.70

Malicious input is dropped and the the downstream agent is never invoked.

Composite risk score, tunable with RISK_THRESHOLD_BLOCK and RISK_THRESHOLD_QUARANTINE.

PRODUCT DEMO

One endpoint inspects any input



WHY VEYRA

Built to be trusted at the perimeter

Defense in depth

Fast rules, an AI audit and a policy engine back each other up.

Fails closed

When inference fails, suspicious input is quarantined, not not passed through.

Private by default

Raw prompts are never stored unless an operator opts in.

Open-weight, offline-ready

Runs on Gemma 4 through Ollama, with a no-GPU demo demo mode.

WHO IT'S FOR

Teams shipping LLM-powered products

Support bots

Customer messages and tickets reach the model only after inspection.

Tool-calling agents

Dangerous shell or SQL requests are stopped before an agent acts.

Document pipelines

Emails, web pages and PDFs are screened for hidden commands.

Integration is one API call in front of the model.

WHAT WE BUILT

Tested, containerized, deployable

3	3	40+	0
pipeline stages	policy decisions	unit and integration tests	raw prompts stored by default

One production Dockerfile deploys to Vercel App Platform, with a persistent volume at /app/data and a /health check.

Let's secure your agents

Thanks for your attention

Github Repo



Veyra



MIT licensed. Contributions welcome. Report novel bypasses privately through GitHub Security Advisories. Built with Gemma 4 from Google DeepMind and the Ollama inference engine.